18 February 1969
SUPERSEDING
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MILITARY SPECIFICATION

SOUP, INSTANT, CREAM OF POTATO AND CREAM OF ONION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers an instantly rehydrating cream of potato and cream of onion soup, for use by the Armed Forces as a component of operational rations.

1.2 Classification.- The product shall be of the following types and classes, as specified (see 6.1):

Type I - Cream of potato.
Type II - Cream of onion.

Class A - Unfortified.
Class B - Fortified.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

EE-S-321 - Shortening Compound and Lard.
EE-S-631 - Spices; Ground and Whole.

FSC 8935

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EE-S-645 - Spice Flavorings; Soluble.
TT-C-495 - Coatings, Exterior, For Tinned Food Cans.
JJJ-O-533 - Onions, Dehydrated.
JJJ-P-630 - Potatoes, White, Dehydrated.
PPP-B-636 - Box, Fiberboard.
PPP-C-29 - Canned Subsistence Items, Packaging and Packing Of.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

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MIL-L-1497 - Labeling of Metal Cans for Subsistence Items.
MIL-S-3271 - Soup and Gravy Base, Beef Flavored.
MIL-M-35052 - Milk, Nonfat, Dry.
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence In.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 101 - Preservation, Packaging and Packing Materials; Test Procedures.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply:

The United States Pharmacopoeial Convention, Inc.

Pharmacopoeia of the United States

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 46 Park Avenue, New York, N. Y. 10016).

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U. S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act
and Regulation Promulgated Thereunder.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402).

Association of Official Agricultural Chemists

Official Methods of Analysis.

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D. C. 20044).

McGraw-Hill Book Co., Inc.

A Dictionary of Color, by A. Maerz and M. Rea Paul.

(Application for copies should be addressed to the McGraw-Hill Book Co., Inc., 330 W. 42nd St., New York, N. Y. 10036).

U. S. Department of Agriculture

Regulations Governing the Meat Inspection
of the U. S. Department of Agriculture.

(Application for copies should be addressed to the Information Division, Consumer and Marketing Service, U. S. Department of Agriculture, Washington, D. C. 20250).

Association of Vitamin Chemists

Methods of Vitamin Assay

(Application for copies should be addressed to the Interscience Publishers, a Division of John Wiley and Sons, Inc., 605 Third Avenue, New York, N. Y., 10016).

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National Research Council

Food Chemicals Codex.

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, Washington, D. C. 20418).

3. REQUIREMENTS

* 3.1 First article (preproduction sample).- When specified (see 6.1), a sample of the finished product shall be submitted to the contracting officer for approval before production is commenced.

3.2 Material.-

* 3.2.1 Potatoes, white, dehydrated.- Dehydrated white potatoes shall meet the requirements for type II, style A of JJJ-P-630.

3.2.2 Dehydrated onions.-

* 3.2.2.1 Onion powder.- Onion powder shall comply with style 2 of JJJ-O-533.

* 3.2.2.2 Onions, minced.- Minced onions shall comply with style 5, white, fancy grade of JJJ-O-533.

* 3.2.2.3 Onions, chopped, toasted.- Toasted chopped onions shall comply with style 1 of JJJ-O-533 except that the onions shall be toasted to a uniformly light brown color with an optical index of 900 to 1700 (see 4.5.2).

* 3.2.3 Nonfat dry milk solids.- Nonfat dry milk solids shall comply with type I, style B, of MIL-M-35052.

3.2.4 Shortening.- Shortening shall comply with type II, class 2 (b) of EE-S-321.

3.2.5 Thickening ingredients.- Any of the thickening ingredients specified in 3.2.5.1 and 3.2.5.2 or 3.2.5.3 may be used, singly or in combination, to result in a soup of smooth consistency, and practically free from any separation of solids within the requirements of 3.3.1.1 or 3.3.1.2, as applicable.

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3.2.5.1 Potatoes, white, dehydrated.- Dehydrated white potatoes, type II, mashed-precooked; rapid rehydrating, style A (granules) or style B (flakes) shall meet the requirements of JJJ-P-630 for the respective styles, except that the particles may be reduced in size to effect thickening.

* 3.2.5.2 Flour.- Flour shall comply with the appropriate definition and standard of identity for flour under the Federal Food, Drug and Cosmetic Act.

* 3.2.5.3 Starch.- Pregelatinized starch which is edible and readily dispersible can be used in quantities which meet the thickening ingredients requirement of 3.3.1.1 or 3.3.1.2 respectively, and the finished product requirement of 3.4.1.

* 3.2.6 Salt.- Salt shall be white, refined sodium chloride, with or without anticaking agent. The salt shall not contain more than 10 parts per million (ppm) of heavy metals of which not more than 2 ppm may be copper. Iodized salt shall not be used.

* 3.2.7 Smoked flavoring.- Smoked flavoring shall consist of debittered yeast, salt, hydrolyzed vegetable protein, or monosodium glutamate, singly or in combination which have been smoked or which contain a smoke flavoring in quantities approved by the Processed Meat Inspection Division, Consumer and Marketing Service, USDA.

3.2.8 Seasoning.-

3.2.8.1 Soluble spice flavorings.- Soluble black pepper and celery spice flavorings shall conform to EE-S-645.

3.2.8.2 Ground spice.- Ground decorticated black pepper and celery seed shall conform to type II of EE-S-631.

* 3.2.9 Hydrolyzed protein.- The hydrolyzed protein shall be obtained by the hydrolysis of any one or any combination of corn, milk, soybean, wheat, or yeast proteins, and it shall have a characteristic odor and flavor.

* 3.2.10 Soup and gravy base beef-flavored.- The beef-flavored soup and gravy base shall meet the requirements of MIL-S-3271.

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* 3.2.11 Vitamin A.- Vitamin A shall be a water-dispersible concentrate of vitamin ester (palmitate) of edible grade. The palmitate shall be refined so that it will not impart any fishy or other objectionable odor or flavor to the product when added at the required level.

* 3.2.12 Thiamine hydrochloride.- Thiamine hydrochloride shall be of a grade equivalent to the Food Chemicals Codex.

* 3.2.13 Ascorbic acid.- Ascorbic acid shall be of a grade equivalent to the Food Chemicals Codex.

3.3 Preparation.-

3.3.1 Formulas.- The soup mixtures shall be prepared in accordance with 3.3.1.1 and 3.3.1.2.

3.3.1.1 Cream of potato soup.-

Ingredients	Percent by weight
Potato granules	51.4
Onions, powdered	1.0
Onions, minced	1.0
Nonfat dry milk solids	20.0
Shortening	12.0
Thickening ingredients	8.0
Salt	4.0
Smoked flavoring	2.0
Black pepper (soluble flavoring) (or equivalent quantity of ground black pepper)	0.3
Celery (soluble flavoring) (or equivalent quantity of ground celery seed)	0.3

* 3.3.1.2 Cream of onion soup.-

Ingredients	Percent by weight
Onions, chopped, toasted	7.5
Onions, minced	5.0
Onions, powdered	8.7
Nonfat dry-milk solids	31.0

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Ingredients	Percent by weight
Salt	5.6
Thickening ingredients	17.0
Shortening	22.5
Hydrolyzed protein	0.6
Soup and gravy base, beef-flavored	1.5
Black pepper (soluble flavoring) (or equivalent quantity of ground black pepper)	0.3
Celery (soluble flavoring) (or equivalent quantity of ground celery seed)	0.3

3.3.2 Processing.- Mixing of components, singly or in combination, prior to final mixing, is permissible. Vitamin fortification may be accomplished by preparing a premix of vitamin A with shortening, and the other vitamins with nonfat dry milk solids.

3.3.3 Blending.- The soup mixtures shall be prepared by blending the ingredients thoroughly in a suitable mixer.

3.4 Finished products.-

* 3.4.1 Rehydration.- The dehydrated soup mix, when added to hot (180° F.) water in the proportion of 1-1/4 ounces (avoirdupois) of dry mix to 8 fluid ounces of water, shall disperse readily and, after stirring with a spoon for not longer than 2 minutes, shall rehydrate into a soup having a smooth consistency with no lumpiness. The color of the prepared soups shall be between A-1 and B-1 when compared with the following plate numbers of Maerz and Paul Dictionary of Color:

Potato soup -- Plate 17.
Onion soup -- Plate 18.

The product shall possess good characteristic flavors and shall be free from rancid, sour, scorched, or any other objectionable flavors.

3.4.2 Moisture content.- The moisture content of the dehydrated soups shall be as follows:

Ingredients	Percent by weight (maximum)
Cream of potato	6.0
Cream of onion	4.5

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* 3.4.3 Vitamin content, when fortified.- One and one-quarter ounces of the dehydrated soup shall contain the following minimum vitamin levels:

Cream of potato

Vitamin A - - - - - 2500 International units (IU's)
Ascorbic acid - - - 37.5 milligrams (mg.)

Cream of onion

Vitamin A - - - - - - - - - - 2500 IU's
Thiamine hydrochloride - - 0.8 mg.
Ascorbic acid - - - - - - - 37.5 mg.

* 3.5 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder

3.6 Workmanship.- The product shall be processed in establishments meeting the requirements of MIL-STD-668. It shall be prepared and processed with minimum delay between the various stages of production.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Preacceptance inspection.- Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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* 4.3.1.1 Examination of ingredients.-

* 4.3.1.1.1 Specification referenced ingredients.- Each specification referenced ingredient (except soluble spice flavoring) shall be examined in accordance with the examination criteria pertinent to the product as contained in the applicable subsidiary specification, except that the sampling plans shall not apply. A sample unit shall be extracted from a primary container. The sample size shall be five sample units. In the event a lot consists of less than five containers, a sample unit shall be extracted from each container in the lot. Quantity of each sample unit and other changes to quality assurance provisions shall be as set forth in table I. One or more defects shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

* Table I.- Specification ingredients

Ingredient	Sample unit
Soup and gravy base	2 ounces
Shortening	4 ounces
Spices, ground (each)	2 ounces 1/
Milk, dry, nonfat	8 ounces
Onions, dehydrated (each style)	4 ounces dehydrated 1/ 2/; sample unit for rehydration as per JJJ-O-533
Potatoes, dehydrated (each type)	8 ounces 3/

1/ Particle size determination shall be performed on a 100 g composite derived from the containers sampled for examination of other product characteristics.

2/ Examination for dark brown or black units shall not be performed on toasted onions.

3/ Examination for particle size shall not be performed.

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* 4.3.1.1.2 Examination of salt, smoked flavoring, starch, soluble spice flavoring, hydrolyzed protein, vitamins A, thiamine hydrochloride and ascorbic acid.- Conformance to identity requirements (e.g., kind, grade, content, etc. as applicable) shall be ascertained by examination of labels, invoices or other valid documents. Nonconformance shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

* 4.3.1.2 Testing of components.- Testing of components described in this specification and changes in the testing of components in subsidiary specifications shall be as shown in table II. Test requirements for characteristics listed below shall be average requirements. A test failure shall indicate an unacceptable component and use of such shall be cause for rejection of the involved end item.

* Table II.- Testing of components.

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref & reqmt para
Cans 1/	1 can w/lid	Cans	S-1	Tin plate	Nearest 0.01 lb. /base box	5.1.1.2 4.5.5
Envelopes	1 yard strip or equivalent in envelopes	Rolls or envelopes	S-2 S-1	Thickness: 2/ Polyethylene	Nearest 0.001 inch	5.1.1.1 4.5.6
				Aluminum foil	Nearest 0.00001 in.	
				Basis wt. paper	Nearest lb/ream	
Salt	1-8 oz. Composite 3/	Pounds	S-2	Heavy metals	Nearest ppm.	3.2.6
				Copper	Nearest ppm.	4.5.1

1/ Tests on can label and external can coating (when required) shall be performed on the same cans submitted for tin plate test.

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2/ In lieu of performing tests, thickness of laminae may be ascertained by examination of pertinent labels, invoices, or other valid documents.

3/ The composite shall be derived from the number of containers indicated by the inspection level.

* 4.3.1.2.1 Testing of toasted chopped onions.- Tests shall be in accordance with the quality assurance provisions of JJJ-O-533 except that the test for optical density shall not apply. In addition, the onions shall be tested for optical index in accordance with 4.5.2 utilizing the same composite sample as required for other tests.

* 4.3.2 Process examination.- Examination shall be performed during formulation of the finished product to determine conformance to 3.3.1. Formulation records shall be maintained. Nonconformance with formulation requirements shall be cause for rejection of the involved finished product.

* 4.3.3 Examination of end item.- Examinations shall be in accordance with tables IV through VIII, and 4.3.3.1 through 4.3.3.4. Sampling plans shall be in accordance with table III. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units for tables IV, V, VII and VIII and percent defective for table VI.

* Table III.- Sampling plans.

Table	Sample unit	Lot size expressed in	Inspection level	AQLs			
				Major A	Major B	Minor	Total
IV	1 filled & closed envelope	Envelopes	I	0.25	1.5	-	4.0
V	1 filled & closed envelope	Envelopes	S-3	-	0.65	-	-
VI	1 filled can or envelope	Cans or envelopes	S-3	-	-	2.5	-

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* TABLE III.- Sampling plans. (Cont'd)

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major A	Major B	Minor	Total
VII	Contents of 1 can or envelope	Cans or envelopes	S-2	-	1.5	-	6.5
VIII	1 filled intermediate carton of envelopes	Cartons	S-3	-	-	2.5	-

* TABLE IV.- Examination of envelopes.

Category			Defect
Major A	Major B	Minor	
101			Tear, hole, or open seal.
	151		Delamination.
	152		Not style or size specified.
		201	Not clean. 1/
	153		Nomenclature or directions for use missing, incorrect, illegible.
		202	Other required label information missing, incorrect or illegible.

1/ Applicable only when not in conflict with regulations referenced in 3.5.

* TABLE V.- Examination of envelopes for leakage. (see 4.5.4)

Category	Defect
Major B	
151	Leakage.
152	Delamination of envelope. 1/

1/ Examine after removal from water.

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TABLE VI.- Examination for net weight. 1/

Category	Defect
Minor	
201	Net weight of envelopes less than 33.6 grams.
202	Net weight of can more than 5 percent under specified weight.

1/ Report results to nearest 0.1 gram for envelopes and nearest 1/4 ounce for cans. Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.

TABLE VII.- Examination of product. 1/

Category		Defect
Major B	Minor	Dehydrated Product
	201	Not thoroughly blended.
		Rehydrated 2/
151		Product does not disperse readily within the prescribed mixing time.
152		Not a smooth soup free of lumps.
	202	Color not as specified.
153		Flavor not characteristic of product.
154		Rancid, sour, scorched or other objectionable flavors.

1/ Presence of foreign material (e.g. filth, insects, glass, paint, metal, etc.) shall be cause for rejection of the lot.

2/ See 3.4.1 for procedure. Make one determination per sample unit.

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* TABLE VIII.- Examination for count of envelopes per carton.

Category	Defect
Minor	
201	More than 2 percent (to nearest unit) under established number of envelopes. 1/
202	Established quantity of envelopes not within required limits.

1/ Lot shall be rejected if sample data indicate a lot average count less than established count.

* 4.3.3.1 Examination of can label.- Examination shall be in accordance with the criteria in MIL-L-1497.

* 4.3.3.2 Examination of exterior can coating.- When exterior can coating is required to be in accordance with TT-C-495, examination shall be in accordance with the criteria in that specification.

* 4.3.3.3 Examination of cans.- Examination of cans for external condition and seam leakage shall be in accordance with the applicable criteria in PPP-C-29.

* 4.3.3.4 Examination of shipping containers.- The filled and closed shipping containers shall be examined in accordance with the criteria in PPP-B-636. In addition, the following defects shall be included in the examination. Major - Markings missing, incorrect or illegible; Minor - Number of cans or intermediate cartons not as specified; Minor - Markings indicate more than 500 envelopes in shipping containers. For level C pack only, the defects applicable to marking and count shall apply. Wire strapped shipping containers intended for transfer at sea shall not be acceptable.

* 4.3.4 Test for oxygen content of cans.- The filled and sealed cans shall be tested for oxygen content in accordance with 4.5.3. The sample unit for testing shall be one filled and sealed can, and the sample size shall be

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the number of cans indicated by inspection level S-2. Lot size shall be expressed in terms of cans. Test requirement shall be applicable to the sample unit. Only cans passing the leakage test shall be submitted for the oxygen test. The AQL, expressed as percent defective shall be 1.5. Results shall be reported to the nearest 0.1 percent and more than 2.0 percent oxygen in a sample unit shall be a defect.

* 4.3.5 Testing of the finished product.- Procedures for testing the finished product for moisture, vitamin A, thiamine hydrochloride and ascorbic acid shall be in accordance with 4.5. The sample for testing shall be a composite of 8 ounces of product derived from the number of cans indicated by inspection level S-2 or the number of envelopes necessary to yield 8 ounces of product. Lot size shall be expressed in primary containers. Results shall be reported to the same decimal place or unit as specified for the test requirement. One or more test failures shall be cause for rejection of the lot.

4.4 Standby test samples.- The Government reserves the right to withdraw and hold standby test samples of components or finished product or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. Samples not used will be returned to the vendor.

4.5 Tests.- Test procedures and controls which differ from those specified herein, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

* 4.5.1 Chemical analyses.- Chemical analyses shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Agricultural Chemists, unless otherwise indicated below:

Test	Source	Method
Salt:		
Copper	Chapter: Metals and Other Elements	Carbamate
Analysis for heavy metals	U. S. Pharmacopoeia, Chapter: General Tests Section: Chemical Tests	Heavy Metals Test

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Test	Source	Method
Finished product:		
Moisture	Chapter: Fruits and Fruit Products	Moisture in Dried Fruits *
Vitamin A	Methods of Vitamin Assay	Carr-Price blue color colorimetric method
Thiamine hydrochloride	Methods of Vitamin Assay	Thiochrome Method
Ascorbic acid	Methods of Vitamin Assay	2,6 dichlorophenol-indophenol photometric method with the Loeffler and Ponting Modification

* Drying time shall be 6 hours at 70° C., and the pressure shall not exceed 50mm. of mercury.

* 4.5.2 Color of chopped onions.- Test for determining the extracted color (optical index) of toasted chopped onions shall be as follows:

Preparation of sample:	Grind sample under low humidity conditions to pass through a 20-mesh screen.
Sample weight:	Use 1 gram of toasted chopped onions.
Procedure:	Mix prepared sample with 100 milliliters (ml) 10 percent sodium chloride solution and allow to stand 15 minutes with occasional stirring. Filter through Whatman No. 12 filter paper, returning filtrate until crystal clear. Determine percent transmission of filtrate at 420 millimicron wave length with a colorimeter or a spectrophotometer standardized at 100 percent transmission with the 10 percent salt solution used for extraction. When instruments with adjustable light transmission slits are used, the narrowest slit width shall be used for greater precision.

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Calculation: Calculate optical density on the basis of 1.0 percent solution, 420-millimicron wave length and 50-millimeters (mm) cell path. Optical density value multiplied by 1000 gives conversion to optical index.

* 4.5.3 Oxygen content.- Oxygen content shall be determined in accordance with method 237 of Fed. Test Method Std. No. 101. When the can has been sealed under reduced pressure, the gases may have to be extracted and compressed prior to testing for oxygen.

4.5.4 Leakage test.- The filled and closed envelope shall be tested for leakage by submerging the container in water, contained in a desiccator or other suitable container, and maintaining a vacuum of 10 inches of mercury (atmospheric pressure 29.9 inches) for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered as signs of leaks.

4.5.5 Tin coating weight.- Tin coating weight shall be determined by any method specified in PPP-C-96.

* 4.5.6 Paper testing.- Basis weight of paper shall be tested in accordance with method 204 of Fed. Test Method Std. No. 101.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- The product shall be packaged in accordance with 5.1.1 or 5.1.2, as specified (see 6.1).

5.1.1 Level A.- The product shall be packaged in accordance with 5.1.1.1 or 5.1.1.2, as specified (see 6.1).

* 5.1.1.1 Envelopes for operational rations.- One and one-quarter ounces of the type I or II product shall be packaged under an atmosphere of nitrogen in a heat-sealable envelope with outside dimensions of 5-1/4 by 4-3/8 inches, ± 1/8 inch in either dimension. All four edges shall be heat sealed. The envelope shall be constructed from material made by hot extruding polyethylene on 0.00035-inch thick aluminum foil, and laminating the plain side of the foil to sulphite or kraft paper. The polyethylene shall have a minimum thickness of 0.001 inch. The paper shall have a basis weight of 25 to 30 pounds per ream (24 by 36 - 500), when tested in accordance with 4.5.6, and shall be processed

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to form a soft pliable sheet. The adhesive for laminating the foil to the paper shall be a suitable odorless and permanently plastic water-resistant compound other than wax. The combined sheet shall show no evidence of delamination when made into bags or heat sealed. The filled and sealed envelopes shall not show leakage or delamination when tested in accordance with 4.5.4. Not less than 24 nor more than 50 envelopes shall be packed on end in an intermediate folding or set-up carton. The number of envelopes per carton shall be the same within the range specified. The oxygen content of the gases in the filled and sealed envelope not less than 2 days after filling, shall not exceed 2.0 percent when tested as specified in 4.5.3.

* 5.1.1.2 Cans.- Type I or II product shall be packaged into an open-top style, round, metal can, with soldered side seam and compound-lined, double-seamed ends. The cans shall be made throughout from not less than commercial 0.25-pound per base box electrolytic tin plate (see 4.5.5), and shall be coated outside with a coating conforming to type I or II, or when specified (see 6.1), type III of TT-C-495. Net contents and can sizes shall be in accordance with table IX, as specified (see 6.1). The product shall be gas-packaged by first removing the air from the filled can and replacing it with nitrogen. If tip soldering is used to close the exhaust vents for nitrogen filling, a suitable noncorrosive flux shall be used. If the ends of the can are clinched prior to vacuumizing and gassing, the flange ends should be indented or dimpled to prevent seal-off trouble during the nitrogen flush. The cans shall be hermetically sealed. The oxygen content of the gases at whatever pressure is in the sealed container, not less than 2 days after filling, shall not exceed 2.0 percent when tested as specified in 4.5.3.

TABLE IX.- Can size, net weight, number per case, and arrangement in shipping container.

Can size	Net weight	Number per case	Arrangement		
			Length	Width	Depth
307 by 409	16-1/4 ounces	24	4	3	2
603 by 700	4 lb. and 12 ounces	6	3	2	1

5.1.2 Level C (cans only).- The product shall be packaged in accordance with 5.1.1.2 except that cans with or without commercial exterior can coating will be acceptable.

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5.2 Packing.- The product, in number and arrangement as specified in table IX, shall be packed in accordance with 5.2.1, 5.2.2, or 5.2.3, as specified (see 6.1).

5.2.1 Level A (cans).- The cans of the product shall be packed on end in a fiberboard box with sleeve, constructed, closed and metal strapped in accordance with style RSC-SL, V2s of PPP-B-636.

5.2.2 Level B (cans).- The cans of the product shall be packed on end in a fiberboard box, constructed and closed in accordance with style RSC, type CF or SF, class domestic of PPP-B-636.

* 5.2.2.1 When specified (see 6.1) the shipping container shall be constructed and closed in accordance with style RSC, V3c or V3s, of PPP-B-636. The shipping container material may also be grade V4s of PPP-F-320. For transfer at sea the shipping container shall be reinforced with other than round wire strapping in accordance with appendix of PPP-B-636.

5.2.3 Level C.-

5.2.3.1 Cans.- The product shall be packed in containers which are acceptable to common carriers for safe transportation to the destination specified in shipping instructions at the lowest transportation rate for such supplies.

5.2.3.2 Envelopes for shipment to ration assembly.- Not more than 500 envelopes in cartons as specified in 5.1.1.1, shall be packed in accordance with 5.2.3.1.

* 5.3 Unit loads.- Soup, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1).

5.4 Labeling and marking.-

* 5.4.1 Envelopes.- The following information shall be printed on the outside of the envelopes in black ink:

NAME OF PRODUCT (in letters larger than any others used on the label).
Net weight
List of ingredients
Name and address of manufacturer
Name and address of packer (if other than manufacturer).

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DIRECTIONS FOR USE

Add contents of package to 1/3 canteen cup (8 fluid ounces) of hot water.
Stir for 1 minute.
Any other information required by the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

5.4.2 Cans.- Cans shall be labeled in accordance with MIL-L-1497. In addition, the following information, as applicable, shall be printed, hand stamped, or lithographed in black letters on the can body or cover:

NOTE: This product is packaged in nitrogen. Distended ends are not necessarily an indication of unsatisfactory product quality.

For 307 by 409 can:

DIRECTIONS FOR USE

13 servings - 1 cup (8 fluid ounces) each.
Stir soup mix to break apart lumps.
Add contents of can slowly (with stirring) to 3-1/4 quarts hot water (near boiling) and stir until smooth.

For 603 by 700 can:

DIRECTIONS FOR USE

60 servings - 1 cup (8 fluid ounces) each.
Stir soup mix to break apart lumps.
Add contents of can slowly (with stirring) to 3-3/4 gallons (15 quarts) hot water (near boiling) and stir until smooth.

5.4.3 Intermediate cartons.- Labeling of intermediate cartons is not required.

5.4.4 Shipping containers.- Shipping containers shall be marked in accordance with MIL-STD-129.

* 5.4.5 Unit loads.- Unit loads shall be marked in accordance with MIL-L-35078 and MIL-STD-129.

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* 6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of product required (see 1.2).
(c) First article (preproduction sample).- When a preproduction sample is required (see 3.1).
(d) Whether preproduction sample is to be furnished (see 3.1).
(e) Levels of packaging and packing required (see 5.1 and 5.2).
(f) Type of packaging required (see 5.1.1).
(g) Size of can packaging required (see 5.1.1.2).
(h) When type III exterior can coating is required (see 5.1.1.2).
(i) When packing specified in 5.2.2.1 is required.
(j) Type and class of unit load required (see 5.3).

6.2 Packing specified in 5.2.2.1 is intended to provide the limited military protection required for: shipment of supplies to certain specified overseas destinations; transfer at sea; and for shipment and storage of Special Program Reserves.

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, SA
Air Force - 45
DSA - SS

Preparing activity:

Army - GL

Project No. 8935-0030

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Soup, Instant, Cream of Potato and Cream of Onion
MIL-S-35056B

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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